

Clause	Purpose
SELECT	Retrieves specific columns
FROM	Specifies the source table
WHERE	Filters rows before grouping
DISTINCT	Removes duplicate values
ORDER BY	Sorts results
GROUP BY	Groups rows with similar values
HAVING	Filters grouped data
LIMIT	Restricts number of rows returned
OFFSET	Skips rows before returning results
INNER JOIN	Returns matching records from both tables
LEFT JOIN	Returns all records from left table and matching from right
RIGHT JOIN	Returns all records from right table and matching from left
FULL OUTER JOIN	Returns all matching and non-matching rows
CROSS JOIN	Returns Cartesian product
UNION	Combines results and removes duplicates
UNION ALL	Combines results and keeps duplicates
IN	Matches multiple values
BETWEEN	Filters within a range
LIKE	Pattern matching
EXISTS	Checks existence of records in subquery
WITH (CTE)	Creates temporary result set
CASE	Implements conditional logic
INSERT INTO	Adds new records
UPDATE	Modifies existing records
DELETE	Removes selected records
TRUNCATE	Removes all rows from a table
CREATE	Creates database objects
ALTER	Modifies database objects
DROP	Deletes database objects

Syntax Example
SELECT name, salary FROM Employees;
SELECT * FROM Employees;
SELECT * FROM Employees WHERE salary > 50000;
SELECT DISTINCT department FROM Employees;
SELECT * FROM Employees ORDER BY salary DESC;
SELECT department, COUNT(*) FROM Employees GROUP BY department;
SELECT department, COUNT(*) FROM Employees GROUP BY department HAVING COUNT(*) > 5;
SELECT * FROM Employees LIMIT 5;
SELECT * FROM Employees LIMIT 5 OFFSET 10;
SELECT * FROM A INNER JOIN B ON A.id=B.id;
SELECT * FROM A LEFT JOIN B ON A.id=B.id;
SELECT * FROM A RIGHT JOIN B ON A.id=B.id;
SELECT * FROM A FULL JOIN B ON A.id=B.id;
SELECT * FROM A CROSS JOIN B;
SELECT city FROM Customers UNION SELECT city FROM Suppliers;
SELECT city FROM Customers UNION ALL SELECT city FROM Suppliers;
SELECT * FROM Employees WHERE dept IN ('IT','HR');
SELECT * FROM Employees WHERE salary BETWEEN 40000 AND 70000;
SELECT * FROM Employees WHERE name LIKE 'J%';
SELECT * FROM Employees WHERE EXISTS (...);
WITH Temp AS (...) SELECT * FROM Temp;
CASE WHEN salary > 50000 THEN 'High' END
INSERT INTO Employees VALUES (...);
UPDATE Employees SET salary=60000 WHERE id=1;
DELETE FROM Employees WHERE id=1;
TRUNCATE TABLE Employees;
CREATE TABLE Employees (...);
ALTER TABLE Employees ADD email VARCHAR(50);
DROP TABLE Employees;

Common Use Case
Fetch required data
Identify data source
Conditional filtering
Unique records
Ranking and sorting
Aggregation
Aggregate filtering
Top N records
Pagination
Combining related data
Find unmatched records
Right-side analysis
Complete comparison
All combinations
Merge datasets
Merge with duplicates
Multiple conditions
Range queries
Search operations
Correlated queries
Complex queries
Categorization
Data insertion
Data modification
Data deletion
Fast table cleanup
Database design
Schema changes
Remove objects